

J MUKHESH

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Three parts of the array

```
import java.util.*;

public class Main{

    public static void main(String args[]){

        Scanner s = new Scanner(System.in);

        int n = s.nextInt();

        long[] nums = new long[n];

        for(int i = 0; i < n; i++){

            nums[i] = s.nextLong();

        }

        long ans = 0, max = 0, smax = 0;

        int i = 0, j = n - 1;

        while(i <= j){

            if(max <= smax){

                max += nums[i];

                i++;

            }else{

                smax += nums[j];

                j--;

            }

            if(max == smax)

                ans = max;

        }

        System.out.print(ans);

    }

}
```

Boat competition

```
import java.util.*;

public class Main{

    public static void main(String args[]){

        Scanner s = new Scanner(System.in);

        int n = s.nextInt();

        for(int i = 0; i < n; i++){

            int m = s.nextInt();

            Map<Integer,Integer> map = new HashMap<>();

            int max = 0;

            for(int j = 0; j < m; j++){

                int cur = s.nextInt();

                map.put(cur,map.getDefault(cur,0) + 1);

                max = Math.max(max,cur);

            }

            int ans = 0;

            for(int j = 2; j <= 2 * max; j++){

                int c = 0;

                for(int k = 1; k <= j / 2; k++){

                    if(!map.containsKey(k) || !map.containsKey(j - k))

                        continue ;

                    if(k == j - k){

                        c += map.get(k) / 2;

                    }else{

                        c += Math.min(map.get(k),map.get(j - k));

                    }

                }

            }

            ans = Math.max(ans,c);

        }

    }

}
```

```
    }  
    System.out.print(ans+" ");  
    }  
    }  
}
```